brms package

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Contents

Introduction

brms: what's that?

Bayesian Regression Models using Stan:
An R package for Bayesian multilevel modelling with the Stan probabilistic programming language, offering:

- Wide range of distributions and link functions
- Flexible model specification compiled during runtime, including crossed and nested parameters.
- Formula language coincides with frequentist `lme4` Package
- User defined priors and covariance structures
- Support for ordinal data, missing or censored data, meta-analysis, multivariate models...

(Bürkner, 2017; Carpenter et al., 2017).

Stan vs. JAGS

- Bayesian analysis of complex models has become easier through the development of MCMC algorithms (e.g. Metropolis-Hastings, Gibbs, Hamiltonian Monte Carlo).
- Major software packages that implement these algorithms: JAGS & Stan.
- JAGS (Just Another Gibbs Sampler) uses Gibbs sampling.
- Stan uses Hamiltonian Monte Carlo & its extension NUTS (No-U-Turn Sampler).
- NUTS has the advantage of converging relatively quickly for a wide range of distributions.

(Bürkner, 2017; Stan Development Team, 2025).

brms, Stan, & c++

- To interface with Stan, brms requires the `rstan` Package.
- brms is more flexible than other packages because it compiles the model during analysis. But this requires a c++ compiler.
  - `Rtools` for Windows
  - `Xcode` for Mac
- After running a model *m*, the underlying Stan code can be accessed via `stancode(m)`

(Bürkner, 2017).

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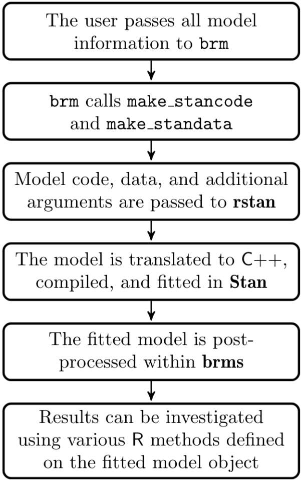
Model fitting procedure with brms



(Bürkner, 2017).

brms tools across the workflow

- Prior predictive checks
- Convergence diagnostics
- Posterior parameter estimates
- Posterior predictive checks
- Model comparison with point estimates
  - Information criteria: WAIC
  - Crossvalidation: LOO
  - Pointwise log-likelihood, R^2
- Model comparison with Bayes factors
  - density ratios
  - bridge sampling

For a complete list see `methods(class = "brmsfit")`.

(Bürkner, 2017).

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Many types of distributions

- Linear & robust linear regression ∼ `gaussian` or `student_t` family with identity link.
- Dichotomous & categorical data ∼ `bernoulli`, `binomial`, or `categorical` families with logit link.
- Count data ∼ poisson, `negbinomial` or `geometric`.
- Response time data, survival regression ∼ `lognormal`, `Gamma`, `exponential`, or `weibull`.
- Ordinal regression ∼ `cumulative`, `cratio`, `sratio`, or `acat`.
- Zero-inflated regression ∼ `zero_inflated_poisson` `zero_inflated_negbinomial`, `zero_inflated_binomial`, `zero_inflated_beta`, `hurdle_poisson`, `hurdle_negbinomial`, or `hurdle_gamma`.

Richard Morey has interactive graphics for different distributions.

(Bürkner, 2017) *Note.* "cratio": continuation ratio, "stratio": stopping ratio, "acat": adjacent category.

Sometimes suboptimal defaults

Every parameter can be given a prior distribution. Parameters for "fixed" effects are called "population level" effects, parameters for "random" effects are called "group level" effects.

- Default for population level effects is a flat improper prior over the reals.
- Defaults for group level effects are set to be weakly informative. The data is used to scale the default "prior" *M* and *SD* for group level effects.
- Alternative priors can be set with the `set_prior()` function.

(Bürkner, 2017).

Formula syntax

Follows format of `lme4` Package

$$\texttt{response} \sim \texttt{pterms} + (\texttt{gterms}|\texttt{group})$$

- `pterms`: population level effects, assumed to be the same across observations.
- `gterms`: group level effects, assumed to vary across the grouping variables specified.

(Bürkner, 2017).

Population level effects

- `m1 <- brm(y ~ 1, data = dat)`
  (intercept-only model)
- `m1 <- brm(y ~ x1 + x2, data = dat)`
  (model with two main effects)
- `m1 <- brm(y ~ x1 + x2 + x1:x2, data = dat)`
  `m2 <- brm(y ~ x1 + x2*x1, data = dat)`
  (model with fixed effects & interaction, m1 = m2)

(Bürkner, 2017, 2018).

Group level effects (1)

- `(1 | g1)`
  (random intercept for group g1)
- `(x1 | g1)`
  (random intercept & random slope for x1 nested in g1)
- `(x1*x2 | g1)`
  (random intercept and random slopes for x1, x2 and the x1:x2 interaction, nested in g1)
- `(0 + x1 | g1)`
  (No random intercept, only random slope for x1 nested in g1)
- `(1 | g1:g2)`
  (intercept nested in new grouping factor that consists of the combined levels of g1 and g2)

(Bürkner, 2017, 2018).

Group level effects (2)

- `(1 | g1 / g2)`
  `(1 | g1) + (1 / g1:g2)`
  (Intercept nested in g1, g1 nested in g2)
- `(x1 || g1)`
  `(0 + x1 | g1) + (1 | g1)`
  (only variances of intercept and x1 but not covariances, i.e. no group level interactions)
- `(1 | g1 + g2)`
  `(1 | g1) + (1 | g2)`
  (crossed random effects: intercept nested in g1 and in g2 - for example trials nested in materials and in participants)

(Bürkner, 2017, 2018).

Formula example

- How long does it take to judge whether a phrase is grammatical in a given language (rt)?
- To what extent does this change on average with phrase length (length) and the presence of subordinate clauses (sub)?
- To what extent does this vary over sentences (mat) and between participants (id)?

```
m1 <- brm(rt ~ length + sub + (1|mat) + (1|id),
          family = lognormal, data = dat)
```